Layer-Importance-Aware AI Model Downloading in Wireless Networks

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Abstract

As AI models proliferate, wireless AI model downloading becomes a key enabler of edge intelligence. However, efficient model delivery over capacity-limited wireless networks remains a fundamental challenge. Recognizing that different model layers contribute unequally to inference accuracy, this paper studies layer-importance-aware multi-user AI model downloading over wireless networks. We first theoretically quantify the accuracy degradation induced by layer-wise transmission distortion, and leverage this metric to formulate accuracy-oriented resource allocation problems under bandwidth, power, and latency constraints. We then develop analog and digital transmission schemes for both AWGN and Rayleigh fading channels. For AWGN channels, the resulting problems for both schemes admit closed-form optimal solutions; for Rayleigh fading, we further design a threshold-based SNR control strategy to handle channel uncertainty, yielding an optimal solution for the analog scheme and a heuristic solution for the digital scheme. Experimental results show that the proposed framework significantly improves inference accuracy over conventional baselines under the same communication budget.

CCS Concepts

• **Networks** → **Network resources allocation**; *Wireless access networks*; • **Computing methodologies** → *Optimization algorithms*; *Neural networks*.

Keywords

AI model downloading, wireless networks, resource allocation, accuracy-oriented transmissions, edge intelligence

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1 Introduction

Traditional cloud AI mainly relies on computation offloading to remote clouds, which may incur excessive latency and raise privacy concerns [17, 29]. These limitations have driven the rise of edge and on-device intelligence [7, 14, 26]. As on-device intelligence proliferates, a fundamental research challenge arises: *how to efficiently download AI models to users over wireless networks*. The urgency of this challenge is underscored by stringent application requirements and the ever-growing size of AI models [5, 33]. For instance, 3GPP Release 18 specifies a 1-second model download requirement for autonomous driving and robotic control, yet vision models used in such applications often exceed 100 MBytes [1]. The challenge is further exacerbated by capacity-limited wireless fading channels, especially for users with unfavorable channel conditions [31]. Traditional error-free transmission combats unreliability via low code rate and retransmissions [2], but such conservative designs conflict with real-time AI service requirements and consume excessive spectrum bandwidth. Fortunately, AI model downloading need *not* be error-free: a model can still retain acceptable performance even when its parameters are corrupted [27], as evidenced by the success of pruning and quantization methods [11, 13]. This observation suggests a paradigm shift in model downloading: optimizing communication resources directly for inference accuracy, rather than minimizing bit errors as in conventional data transmission [24, 25].

However, despite extensive research on communication efficiency in edge AI, efficient wireless model downloading remains largely unexplored. On the communication side, existing edge-AI resource allocation schemes often target feature or gradient transmission and do not address how limited wireless resources should be allocated to *pre-trained model parameters* to preserve *inference accuracy*. On the machine learning side, existing studies typically quantify the effect of parameter distortion through indirect surrogates such as logit space perturbation, without explicitly characterizing its impact on *inference accuracy* (e.g., top-1 accuracy). Thus, directly adopting these surrogates as optimization objectives will result in suboptimal resource allocation. Moreover, even these studies consider only the effect of parameter distortion (e.g., quantization), rather than connecting communication quantities, such

as signal-to-noise ratio (SNR) or bit error probability, to inference quality. Consequently, two fundamental questions remain open:

Q1: *Can the relationship between transmission distortion and inference accuracy be characterized, so that model downloading can be optimized directly with respect to inference accuracy?*

Q2: *Based on such a characterization, how should wireless resources be optimally allocated to maximize inference accuracy for AI model downloading?*

To answer these two questions, this paper studies layer-importance-aware AI model downloading in wireless networks. The key insight is that different layers can contribute very unequally to the final model performance [4, 19, 34], and thus wireless resources should be preferentially invested in the layers that matter most. To the best of our knowledge, this paper is the first to explicitly characterize the effects of layer-wise transmission distortion on inference accuracy (i.e., **Q1**) and to develop an accuracy-oriented resource allocation framework for model downloading (i.e., **Q2**). Specifically, we first establish a layer-wise metric for accuracy degradation under both analog and digital transmission schemes. Based on this formulation, we develop accuracy-oriented resource allocation schemes for both analog and digital transmissions under AWGN and Rayleigh fading channels subject to bandwidth, power, and deadline constraints. The main contributions of this paper are summarized as follows.

- We propose a layer-importance-aware model downloading framework in multi-user systems, formulated as an accuracy degradation minimization problem under latency, bandwidth, and average-power constraints.
- We analytically characterize the impact of layer-wise transmission distortions on inference accuracy by deriving a tractable layer-wise accuracy degradation metric.
- Based on the derived accuracy metric, we instantiate the proposed framework with analog and digital transmission schemes over AWGN channels. For both cases, the original problems are equivalently transformed into convex formulations, and the closed-form optimal solutions are derived.
- We further extend the framework to Rayleigh fading channels via a threshold-based SNR control strategy that efficiently handles channel dynamics. Under this setting, we develop an optimal solution for analog schemes and a heuristic solution for digital schemes.
- Simulation results show that the proposed framework significantly improves inference accuracy over conventional baselines, with particularly large gains under resource-constrained scenarios (e.g., up to 39.3% accuracy improvement under the same communication budget). The results also suggest that analog schemes generally outperform digital schemes for model downloading in resource-constrained regimes.

2 Related Work

Communication-efficient Edge Intelligence. A number of communication paradigms have been developed for edge intelligence, including federated learning [20], split learning [9], split inference [16, 17], and task-oriented communication [24, 25, 30]. A common theme of these works is that communications should be co-designed with underlying learning or inference objectives, rather than optimized purely for conventional metrics such as rate or bit-error probability. Despite this, most of these studies mainly focus on the exchange of inference- or training-related information, such as intermediate features, gradients, or model updates, rather than on the direct delivery of pre-trained AI models for *inference accuracy*.

Wireless AI Model Delivery. Prior works have investigated AI model delivery in wireless and edge networks from different perspectives, including model caching and model downloading [22]. Edge model caching reduces access latency by proactively storing popular models at edge servers [23], while model downloading schemes aim to reduce latency and bandwidth costs upon model delivery [12, 31]. The most related work is [35], in which Zhou et al. designed a parameter-sensitivity-aware retransmission protocol for AI model downloading, demonstrating that delivery efficiency can be improved by treating parameter packets differently according to their importance. Nevertheless, this work considers only packet-level retransmissions for a single user; it neither formulates a multi-user wireless resource allocation problem nor explicitly links transmission distortion to final inference accuracy.

Analysis of Model Parameter Sensitivity. Another relevant line of research concerns the non-uniform sensitivity of neural networks. Prior studies have shown that different layers or parameters can contribute unequally to model performance [4, 19, 34]. Related observations also underlie many parameter-importance-aware pruning and quantization methods [11, 13].

While these results offer valuable engineering insights, existing work in this area focuses on model compression, quantization, or efficient inference rather than on wireless model delivery. As a result, they do not consider stochastic parameter corruptions induced by wireless channels. Moreover, to quantify the impact of parameter distortion, these studies typically rely on indirect mathematical surrogates, such as logit space perturbations [34], rather than explicitly characterizing the resulting degradation in final inference accuracy.

3 System Model and Problem Formulation

In this section, we present our system model and then formulate the problem.

3.1 System Model

Consider a wireless network consisting of one BS (base station) and m UEs (user equipments), as shown in Fig. 1. Each UE requests to download an AI model from the BS for its own task execution. BS serves multiple UEs simultaneously, each of which may request different models.[1] We denote the number of layers in the model requested by the i-th UE as D_i, and the number of parameters in its j-th layer as $N_{i,j}$. To characterize heterogeneous latency requirements, let t_i denote the transmission deadline of the i-th UE. Without loss of generality, the UEs are indexed in ascending order of their deadlines, i.e., $t_1 \leq t_2 \leq \cdots \leq t_m$.

Due to the stringent latency and bandwidth constraints, it is generally infeasible to ensure error-free model transmission for all UEs. Therefore, instead of pursuing error-free delivery, the BS needs to judiciously allocate limited communication resources to control

[1] Our scheme can also be extended to the download of only a portion of a model, i.e., a subset of its parameters.

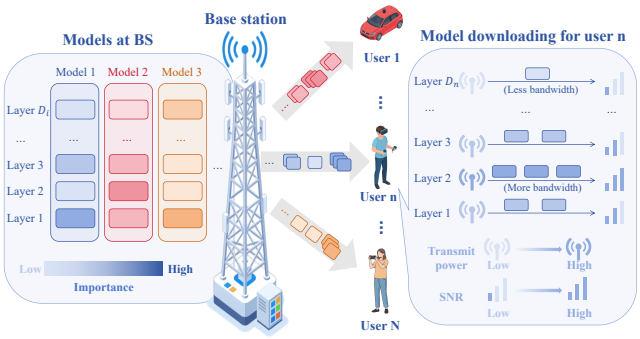


Figure 1: The proposed model downloading framework, which judiciously determines transmit power and bandwidth allocation according to layer importance.

model accuracy, taking into account the importance of model layers. For practical implementations and performance comparison, we study both *analog* and *digital* transmission schemes. Analog transmission has been shown to be highly effective for machine learning tasks, which are inherently tolerant to transmission errors and noise [18]. Digital transmission, by contrast, is more compatible with existing mobile networks. Let $c_{i,j}$ denote the average number of real-valued symbols required for one transmission of one parameter in layer (i, j) (the j-th layer of user i). Since one complex symbol contains two real-valued dimensions, the symbol budget over duration t and bandwidth B is $2Bt$. Accordingly, $c_{i,j} = 1$ for analog transmission, while for digital transmission with $b_{i,j}$-bit quantization and $M_{i,j}$-QAM, $c_{i,j} = \frac{2b_{i,j}}{\log_2(M_{i,j})}$.

3.2 Problem Formulation

Based on the system model, we now formulate a unified resource allocation problem for multi-user model downloading, where the BS jointly allocates bandwidth and transmit power to minimize weighted accuracy degradation. Let $B_{i,j,k}$ denote the bandwidth allocated to layer (i, j) during interval $(t_{k-1}, t_k]$, with $\tau_k = t_k - t_{k-1}$ being the span of the k-th interval. To enhance the received quality of layer (i, j), the BS can allocate additional resource to transmit it repeatedly. Under ideal Nyquist signaling, the equivalent average number of transmissions for layer (i, j) can be expressed as:

$$T_{i,j} = \frac{1}{c_{i,j} N_{i,j}} \sum_{k=1}^{i} 2B_{i,j,k}\tau_k, \tag{1}$$

where $c_{i,j}$ depends on the mode (digital or analog transmissions), as detailed previously. Here, the summation upper limit is i because the i-th UE must complete model delivery before deadline t_i, restricting layer (i, j) to occupy bandwidth within the first i intervals. Furthermore, every parameter of layer (i, j) must be delivered at least once, which introduces the following constraint:

$$T_{i,j} \geq 1, \qquad \forall i = 1, \ldots, m, ; j = 1, \ldots, D_i. \tag{2}$$

Since the system bandwidth is limited to B, the bandwidth allocated in each interval cannot exceed the available spectrum, i.e.,

$$\sum_{i=k}^{m} \sum_{j=1}^{D_i} B_{i,j,k} \leq B, \qquad \forall k = 1, \ldots, m, \tag{3}$$

where the summation starts from $i = k$ as the layers whose deadlines fall at or after t_k can still be scheduled in interval k.

Moreover, the base station is subject to an average power constraint. Since layer (i, j) occupies $\sum_{k=1}^{i} 2B_{i,j,k}\tau_k$ symbols in total, the corresponding average power constraint is given by

$$\sum_{i=1}^{m} \sum_{j=1}^{D_i} \sum_{k=1}^{i} 2B_{i,j,k}\tau_k P_{i,j} \leq \sum_{i=1}^{m} \sum_{j=1}^{D_i} \sum_{k=1}^{i} 2B_{i,j,k}\tau_k \bar{P}, \tag{4}$$

where $P_{i,j}$ is the transmit power allocated to layer (i, j), and $\bar{P}$ is the average power budget.

Finally, we formulate the optimization problem of bandwidth and power allocation to minimize the weighted total accuracy degradation under resource and latency constraints:

$$(P1) \quad \begin{aligned} \min_{\mathbf{B},\mathbf{P}} \quad & \frac{\sum_{i=1}^{m} r_i \, \Delta acc_i(\mathbf{B}_i, \mathbf{P}_i)}{\sum_{i=1}^{m} r_i} \\ \text{s.t.} \quad & (1) - (4) \\ & B_{i,j,k} \geq 0, \; P_{i,j} \geq 0, \end{aligned} \tag{5}$$

where r_i denotes the weighting factor of the i-th UE, $\mathbf{B}$ and $\mathbf{P}$ collect all bandwidth and power allocation variables, respectively. $\Delta acc_i(\mathbf{B}_i, \mathbf{P}_i)$ denotes the resulting accuracy degradation (compared with error-free transmissions) of the i-th UE under the adopted bandwidth and power allocation.

To optimize (P1), we must derive the expression of $\Delta acc_i(\mathbf{B}_i, \mathbf{P}_i)$ by understanding how errors in layer-specific parameters influence inference accuracy, which will be developed in the next section.

4 Characterization of Accuracy Degradation

In this section, we will investigate the relationship between parameter perturbations and final prediction quality, thereby providing an analytical expression of $\Delta acc_i(\mathbf{B}_i, \mathbf{P}_i)$ in (P1). Prior work [34] only analyzes distortion in the logit space, without providing a direct expression of final inference accuracy. Moreover, their expression is not linked to transmission quality. To address this gap, we will directly show the effects of parameter transmission distortion on final inference accuracy.

4.1 Theoretical Analysis

Consider a DNN with D layers and parameter set $\mathbf{w} = \{w_1, w_2, \ldots, w_D\}$, where $w_i \in \mathbb{R}^{d_i}$ denotes the parameter vector of the i-th layer. Let $\Delta w_i \in \mathbb{R}^{d_i}$ be the perturbation on the i-th layer parameters, and $f(x; \mathbf{w}) \in \mathbb{R}^F$ denote the output logits for an input sample x. For notational simplicity, we define

$$f_0 \triangleq f(x; \mathbf{w}). \tag{6}$$

The perturbed logits can be expressed by the second-order Taylor expansion with higher-order terms

$$\hat{f} = f_0 + \sum_{i=1}^{D} J_i \Delta w_i + \frac{1}{2} \sum_{i=1}^{D} \sum_{j=1}^{D} \Delta w_i^T G_{i,j} \Delta w_j + o(\|\Delta w\|^2), \tag{7}$$

where $J_i = \partial f / \partial w_i \in \mathbb{R}^{F \times d_i}$ is the Jacobian from the i-th layer parameters to the logits, and $G_{i,j}$ denotes the corresponding second-order derivative block. The cross term $\Delta w_i^T G_{i,j} \Delta w_j$ for $i \neq j$ characterizes the interaction between perturbations from different layers in the nonlinear propagation of the network, while the case $i = j$

corresponds to the quadratic self-interaction of the i-th layer perturbation.

Since the classification accuracy (e.g., top-1 accuracy) is not differentiable, we introduce a locally smoothed surrogate response $A(\cdot)$ mapped from the logit space. The function $A(\cdot)$ is used only for local sensitivity analysis and for motivating the proposed metric; the corresponding coefficients are estimated empirically in practice. Define the accuracy variation induced by perturbations as $\Delta A \triangleq A(\hat{f}) - A(f_0)$. Let $\nabla A(f_0)$ and H denote the gradient and Hessian of $A(\cdot)$ with respect to the logits, both evaluated at f_0, respectively. We make the following assumptions to derive the accuracy [15, 32].

Assumption 1. *The performance response $A(\cdot)$ is twice continuously differentiable in a neighborhood of f_0.*

Assumption 2. *The perturbation in each layer is sufficiently small compared with the original model parameters.*

Assumption 3. *The perturbations in different layers are zero-mean and mutually independent.*

Under Assumptions 1–3, the expected accuracy degradation admits the following second-order approximation.

Theorem 1 (Second-order approximation of expected accuracy degradation). *Under Assumptions 1–3, the expected accuracy degradation can be approximated as*

$$E[\Delta A] = \frac{1}{2}\sum_{i=1}^{D} \mathrm{tr}\Big(\Big(M_i + J_i^T H J_i\Big)\Sigma_i\Big), \tag{8}$$

where

$$\Sigma_i \triangleq E[\Delta w_i \Delta w_i^T] \tag{9}$$

is the perturbation covariance of the i-th layer, and

$$M_i \triangleq \sum_{k=1}^{F} \nabla A_{(k)}\, G_{i,i}^{(k)} \tag{10}$$

collects the second-order self-interaction terms caused by the nonlinear mapping from the i-th layer parameters to the logits.

Proof. See Appendix A in technical report [8]. □

Theorem 1 shows that the effect of layer-wise noise contains two additive components. The term $J_i^T H J_i$ captures how logit perturbation affects the final performance response, while M_i reflects the additional contribution introduced by the nonlinear propagation of the neural network itself. For a fully linear network, the latter term vanishes, i.e., $M_i = 0$.

Inspired by the notion of signal-to-noise ratio in communication systems, we define the layer signal-to-noise ratio (LSNR) as follows. This metric offers a convenient means of mapping communication quality to the final accuracy, as detailed in Subsection 4.2.

Definition 1 (Layer signal-to-noise ratio). *The layer signal-to-noise ratio (LSNR) of the i-th layer is defined as*

$$LSNR_i \triangleq \frac{E[\|w_i\|^2]}{E[\|\Delta w_i\|^2]} = \frac{E[\|w_i\|^2]}{\mathrm{tr}(\Sigma_i)}. \tag{11}$$

Based on Theorem 1, the expected accuracy degradation can be further rewritten into a layer-wise additive form.

Corollary 1 (Layer-wise additive approximation). *Under Assumptions 1–3, the expected accuracy degradation can be approximately expressed as*

$$E[\Delta A] = \sum_{i=1}^{D} \beta_i \alpha_i \frac{1}{LSNR_i}, \tag{12}$$

where

$$\beta_i \triangleq \frac{\mathrm{tr}\big((M_i + J_i^T H J_i)\Sigma_i\big)}{2\mathrm{tr}(J_i^T J_i \Sigma_i)}, \tag{13}$$

is the sensitivity coefficient of the i-th layer, and

$$\alpha_i \triangleq \frac{\mathrm{tr}(J_i^T J_i \Sigma_i)\, E[\|w_i\|^2]}{\mathrm{tr}(\Sigma_i)}, \tag{14}$$

is the perturbation propagation coefficient.

Proof. Starting from Theorem 1, we have

$$E[\Delta A] = \frac{1}{2}\sum_{i=1}^{D} \mathrm{tr}\Big((M_i + J_i^T H J_i)\Sigma_i\Big). \tag{15}$$

For each layer, multiply and divide the i-th term by $\mathrm{tr}(J_i^T J_i \Sigma_i)$ and $\mathrm{tr}(\Sigma_i)$ to obtain

$$E[\Delta A] = \frac{1}{2}\sum_{i=1}^{D} \frac{\mathrm{tr}\big((M_i + J_i^T H J_i)\Sigma_i\big)}{\mathrm{tr}(J_i^T J_i \Sigma_i)} \cdot \frac{\mathrm{tr}(J_i^T J_i \Sigma_i)}{\mathrm{tr}(\Sigma_i)} \cdot \mathrm{tr}(\Sigma_\mathrm{i}). \tag{16}$$

Using the definition of $LSNR_i$ in (11), we further derive

$$\mathrm{tr}(\Sigma_i) = E[\|w_i\|^2]\frac{1}{LSNR_i}. \tag{17}$$

Substituting the above relation into the previous expression yields

$$E[\Delta A] = \sum_{i=1}^{D} \beta_i \alpha_i \frac{1}{LSNR_i}, \tag{18}$$

where β_i and α_i are defined in (13) and (14), respectively. This completes the proof. □

Corollary 1 provides a compact interpretation of the expected performance degradation. Specifically, β_i characterizes the sensitivity of the performance response to unit logit perturbation energy, while α_i measures how efficiently the perturbation injected into the i-th layer propagates to the logit space. Therefore, the expected performance degradation is jointly determined by the layer sensitivity β_i, the perturbation propagation coefficient α_i, and the layer-wise noise level measured by $LSNR_i$.

It is worth noting that both β_i and α_i are invariant to a uniform scaling of the perturbation covariance, provided that the covariance shape remains unchanged and only the perturbation amplitude varies. Therefore, for a fixed perturbation pattern, these two coefficients mainly characterize the intrinsic sensitivity of the model, while the noise magnitude is explicitly captured by $LSNR_i$.

4.2 Mapping from Communication Quality to Inference Accuracy

Based on the previously derived accuracy, this subsection establishes how communication quality is mapped to the corresponding inverse LSNR and thus inference accuracy under different transmission modes. Since inference accuracy depends on LSNR according to Corollary 1, we will focus on mapping SNR to LSNR below.

4.2.1 Analog Transmissions. For analog transmission, the resulting LSNR is determined by the effective received SNR after retransmission and combining. For the j-th layer of the i-th UE, we use $\gamma_{i,j}^{\text{tot}}$ to denote the effective SNR of one parameter. In general, $\gamma_{i,j}^{\text{tot}}$ is a random variable with a certain distribution; it reduces to a constant only in special cases such as deterministic AWGN channels.

Proposition 1 (Analog communication-to-LSNR mapping). *For analog transmission, the inverse LSNR of the j-th layer of the i-th UE is given by*

$$\frac{1}{LSNR_{i,j}} = \mathbb{E}\left[\frac{1}{\gamma_{i,j}^{\text{tot}}}\right], \tag{19}$$

where the expectation is taken over the distribution of the effective combined SNR $\gamma_{i,j}^{\text{tot}}$.

Proof. See Appendix B in the technical report [8]. □

Proposition 1 shows that, for analog transmission, the communication-to-LSNR mapping is completely determined by the distribution of the effective combined SNR at the receiver side.

4.2.2 Digital Transmissions. Consider a parameter uniformly quantized into a b-bit representation over the interval $[w_{\min}, w_{\max}]$. Let $q \in \{0, 1, \ldots, 2^b - 1\}$ denote its quantization index, which can be written as

$$q = \sum_{k=0}^{b-1} n_k 2^k, \tag{20}$$

where $n_k \in \{0, 1\}$ denotes the k-th bit, and $k = 0$ corresponds to the least significant bit. Let

$$\Delta = \frac{w_{\max} - w_{\min}}{2^b - 1} \tag{21}$$

denote the quantization step size. Then, the parameter distortion can be directly computed from bit errors as follows.

Proposition 2 (Digital communication-to-LSNR mapping). *Consider the j-th layer of the i-th UE containing $N_{i,j}$ quantized parameters. Suppose that all transmitted bits are independently flipped with identical bit-error probability $P_{b,i,j}$. Then the inverse LSNR of this layer can be approximated as*

$$\frac{1}{LSNR_{i,j}} \approx \zeta_{i,j} P_{b,i,j}, \tag{22}$$

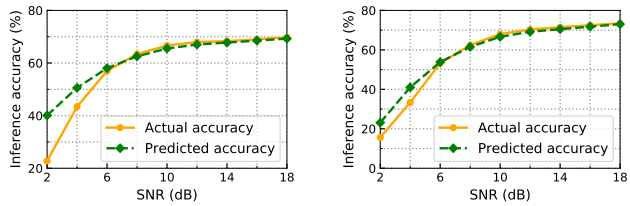
where

$$\zeta_{i,j} = \frac{N_{i,j}\Delta^2 \sum_{k=0}^{b-1} 4^k}{\mathbb{E}\left[\|\mathbf{w}_{i,j}\|^2\right]} \tag{23}$$

is a layer-dependent coefficient determined by the quantization interval and the average parameter energy of that layer.

Proof. See Appendix C in the technical report [8]. □

Proposition 2 indicates that, for digital transmission, the LSNR is characterized through the bit-error probability. By incorporating the expressions of layer-wise inverse LSNR metrics into Corollary 1, we can quantify the accuracy degradation under both analog and digital transmission schemes.



(a) ResNet-18 on ImageNet **(b) MobileNet on CIFAR-100**

Figure 2: Comparison between the predicted and actual accuracies under different SNR. The prediction matches the actual accuracy well in the high-SNR regime, while a gradually increasing deviation is observed as the SNR decreases.

4.3 Empirical Validation

In practice, we cannot explicitly compute α_i and β_i based on J_i, $G_{i,j}$, or H. Instead, α_i and β_i are obtained empirically. Specifically, to estimate α_i, random perturbation is injected into the i-th layer under a prescribed low-noise regime, and the resulting logit perturbation energy is measured. To estimate β_i, the corresponding performance degradation and the logit perturbation energy are measured, and their ratio is used to characterize the local sensitivity. To ensure the validity of the second-order approximation, the perturbation level should remain sufficiently small. Once the layer-wise noise levels are known, Corollary 1 can be used to predict the inference performance of the AI model. Details of the empirical estimation are provided in Appendix D of the technical report [8].

Fig. 2 compares the predicted accuracy with the actual accuracy under different noise levels. It is observed that the predicted accuracy matches the actual accuracy well in the high-SNR regime, which validates the effectiveness of the proposed metric. As the noise level increases, the discrepancy gradually becomes visible, since higher-order terms are no longer negligible. Fortunately, considering that effective SNR is often greater than 10 dB in practical wireless systems, our theoretical analysis provides good approximations in most cases [10]. In this SNR range, the prediction error remains below 2% across both models.

5 Resource Allocation over AWGN Channels

Building on the accuracy degradation in Section 4, this section develops the resource allocation strategies for both analog and digital transmissions over static AWGN channels. Under such channel conditions, each problem admits an optimal solution.

5.1 Analog Transmissions

Let h_i and σ_i^2 denote the channel gain and noise variance of the i-th UE, respectively. With retransmission and combining, the effective received SNR of layer (i, j) is deterministic and given by

$$\gamma_{i,j}^{\text{tot}} = \frac{h_i^2 T_{i,j} P_{i,j}}{\sigma_i^2}. \tag{24}$$

where $T_{i,j}$ can be expressed through (1). By Proposition 1, the corresponding inverse LSNR is

$$\frac{1}{LSNR_{i,j}} = \frac{\sigma_i^2 c_{i,j} N_{i,j}}{h_i^2 P_{i,j} \sum_{k=1}^{i} 2B_{i,j,k}\tau_k}. \tag{25}$$

Substituting (25) into the weighted accuracy-loss model in P1 yields the following optimization problem:

$$(P2) \quad \min_{\mathbf{B},\mathbf{P}} \quad \frac{1}{\sum_{i=1}^{m} r_i} \sum_{i=1}^{m} r_i \sum_{j=1}^{D_i} \beta_{i,j}\alpha_{i,j} \frac{\sigma_i^2 c_{i,j} N_{i,j}}{h_i^2 P_{i,j} \sum_{k=1}^{i} 2B_{i,j,k}\tau_k}$$
$$\text{s.t.} \quad (1)-(5).$$

The product of $P_{i,j}$ and $B_{i,j,k}$ introduces nonconvexity into the problem. To address this issue, we introduce the auxiliary variable

$$z_{i,j} \triangleq \frac{P_{i,j}}{c_{i,j}N_{i,j}} \sum_{k=1}^{i} 2B_{i,j,k}\tau_k, \tag{26}$$

which captures the coupled effect of power and bandwidth allocation for layer (i, j). The original problem could be reformulated into an equivalent convex optimization problem. As a result, the closed-form optimal solution can be obtained as follows.

Theorem 2 (Closed-form for analog schemes under AWGN channels). *For P2, the optimal allocation can be obtained from*

$$\frac{P_{i,j}^*}{c_{i,j}N_{i,j}} \sum_{k=1}^{i} 2B_{i,j,k}^*\tau_k = \frac{2\bar{P}Bt_m\, a_{i,j}}{\sum_{i=1}^{m}\sum_{j=1}^{D_i} a_{i,j}c_{i,j}N_{i,j}}. \tag{27}$$

with

$$a_{i,j} = \sqrt{\frac{r_i\beta_{i,j}\alpha_{i,j}\sigma_i^2}{h_i^2 c_{i,j}N_{i,j}}}. \tag{28}$$

Accordingly, the resulting accuracy degradation of the i-th UE is

$$\Delta acc_i = \frac{\sum_{i=1}^{m}\sum_{j=1}^{D_i} a_{i,j}c_{i,j}N_{i,j}}{2\bar{P}Bt_m r_i} \sum_{j=1}^{D_i} a_{i,j}c_{i,j}N_{i,j}, \tag{29}$$

and the overall weighted accuracy degradation is

$$\Delta acc = \frac{\left(\sum_{i=1}^{m}\sum_{j=1}^{D_i} a_{i,j}c_{i,j}N_{i,j}\right)^2}{2\bar{P}Bt_m \sum_{i=1}^{m} r_i}. \tag{30}$$

Proof. The proof can be found in Appendix E in the technical report [8]. □

Although the bandwidth and power constraints are formulated as inequalities, they are active at the optimum; otherwise, the unused resource can be assigned to any layer with nonzero sensitivity to further reduce the objective, which contradicts optimality. The same argument applies to the subsequent formulations.

Insights: Theorem 2 shows that the product of transmit power and the equivalent number of transmissions is allocated proportionally to $a_{i,j}$. Hence, more effective transmission energy is assigned to layers with higher sensitivity, larger parameter sizes, or poorer channel conditions. Moreover, the weighted accuracy degradation scales as $(\bar{P}Bt_m)^{-1}$, showing the impact of resource and deadline budgets.

5.2 Digital Transmissions

For mobile networks, we often employ discrete quantization and modulation. Specifically, model parameters are first quantized and then modulated using M-QAM. Under AWGN channels, the bit-error rate (BER) of M-QAM can be approximated by [6]

$$P_{b,i,j} \approx \frac{4}{\log_2(M_{i,j})}\left(1-\frac{1}{\sqrt{M_{i,j}}}\right) Q\left(\sqrt{\frac{3\gamma_{i,j}^{\text{tot}}}{M_{i,j}-1}}\right), \tag{31}$$

where $\gamma_{i,j}^{\text{tot}}$ denotes the effective received SNR after retransmission and combining. Then (31) can be rewritten as

$$P_{b,i,j} \approx K_{i,j} Q\left(\sqrt{\eta_{i,j}\frac{h_i^2 T_{i,j}P_{i,j}}{\sigma_i^2}}\right), \tag{32}$$

where

$$K_{i,j} = \frac{4}{\log_2(M_{i,j})}\left(1-\frac{1}{\sqrt{M_{i,j}}}\right), \tag{33}$$

$$\eta_{i,j} = \frac{3}{M_{i,j}-1}. \tag{34}$$

By Proposition 2, the inverse LSNR is approximately proportional to the BER. Therefore, the weighted accuracy-loss minimization problem under AWGN channels can be written as

$$(P3) \quad \min_{\mathbf{B},\mathbf{P}} \quad \frac{1}{\sum_{i=1}^{m} r_i} \sum_{i=1}^{m} r_i \sum_{j=1}^{D_i} \beta_{i,j}\alpha_{i,j}\zeta_{i,j}K_{i,j} Q\left(\sqrt{\eta_{i,j}\frac{h_i^2 P_{i,j}\sum_k 2B_{i,j,k}\tau_k}{\sigma_i^2 c_{i,j}N_{i,j}}}\right)$$
$$\text{s.t.} \quad (1)-(5).$$

Following the same transformation procedure as for P2, the semi-closed-form solution to P3 can be obtained as follows.

Theorem 3 (Semi-closed-form for digital schemes under AWGN channels). *For the digital AWGN case, the problem admits a semi-closed-form optimal solution. The optimal effective-energy allocation is*

$$\frac{P_{i,j}^*}{c_{i,j}N_{i,j}} \sum_{k=1}^{i} 2B_{i,j,k}^*\tau_k = \frac{1}{e_{i,j}} W\left(\frac{d_{i,j}^2 e_{i,j}^2}{8\pi(\lambda^*)^2}\right), \tag{35}$$

where $d_{i,j} = \frac{r_i\beta_{i,j}\alpha_{i,j}\zeta_{i,j}K_{i,j}}{c_{i,j}N_{i,j}}$, $e_{i,j} = \frac{\eta_{i,j}h_i^2}{\sigma_i^2}$. $W(\cdot)$ *is the Lambert-W function, and* $\lambda^* > 0$ *is the dual variable associated with the average-power constraint. Here,* λ^* *is a scalar dual variable, which can be efficiently obtained by a simple one-dimensional search.*

Proof. The proof can be found in Appendix F in the technical report [8]. □

Insights: Theorem 3 suggests that, compared with the analog solution, the digital allocation is less aggressively amplified by large inter-layer disparities. Mathematically, this is because the Lambert-W function grows slowly for large arguments, so the resource assigned to highly dominant layers increases more moderately. Intuitively, this reflects the "cliff effect" of digital transmissions: once a layer receives sufficient resources to achieve a small bit error rate, allocating additional resources yields diminishing returns.

Overall, these AWGN-channel-based solutions illustrate how to allocate resources when channels remain stable. Yet, to apply this framework in dynamic edge AI settings, we also need to adjust these strategies to cope with channel variability, as developed in the next section.

6 Resource Allocation over Rayleigh Channels

In the previous section, we established the optimal resource allocation strategies under static AWGN channels. By taking channel dynamics into account, this section extends our proposed framework to Rayleigh fading channels. We will develop an optimal solution for analog transmissions and a heuristic solution for digital transmissions.

Different from the static-channel case, the instantaneous channel gain varies during transmission, which leads to random fluctuations in the received SNR. To simplify transmitter control, we adopt a fixed transmit-power strategy and regulate the received quality through a threshold-based stopping rule [3]. Specifically, for each layer (i, j), a target received SNR threshold $\Gamma_{i,j}$ is preset at the receiver. Transmission continues until the accumulated received SNR reaches $\Gamma_{i,j}$, whereupon the receiver stops further retransmissions for that parameter.

6.1 Analog Transmissions

Under Rayleigh fading and the threshold-based stopping rule, we can derive LSNR below, which serves as an essential step to quantify the accuracy degradation.

Proposition 3 (Effective layer SNR for analog transmission under Rayleigh channels). *For analog transmission, the effective layer SNR under the stopping rule is characterized through the average reciprocal combined SNR. Specifically, the layer SNR satisfies*

$$\frac{1}{LSNR_{i,j}} = \frac{1}{\bar{\gamma}_{i,j}} e^{\frac{\Gamma_{i,j}}{\bar{\gamma}_{i,j}}} E_1\left(\frac{\Gamma_{i,j}}{\bar{\gamma}_{i,j}}\right), \tag{36}$$

where $E_1(\cdot)$ denotes the exponential integral function.

Proof. The detailed proof is deferred to Appendix G in technical report [8]. □

Similarly, based on the derived LSNR, accuracy degradation from analog transmissions under Rayleigh fading can be computed through Corollary 1. To this end, the analog resource-allocation problem over Rayleigh fading channels is formulated as

$$(P4) \quad \min_{\mathbf{B},\mathbf{P}} \quad \frac{1}{\sum_{i=1}^m r_i} \sum_{i=1}^m r_i \sum_{j=1}^{D_i} \beta_{i,j}\alpha_{i,j} \frac{\sigma_i^2}{\sigma_{h_i}^2 T_{i,j} P_{i,j}} G(T_{i,j}) \quad \text{s.t.} \quad (1)-(5),$$

where $G(x) = xe^{x-1}E_1(x-1)$. To maintain consistency with the optimization variables introduced earlier, $\bar{\gamma}_{i,j}$ and $\Gamma_{i,j}$ are reformulated here as functions of $B_{i,j,k}$ and $P_{i,j}$. Specifically, under the stopping rule, the expected number of transmissions is $T_{i,j} = 1 + \Gamma_{i,j}/\bar{\gamma}_{i,j}$ (determined by the bandwidth allocation in (1)), and $\bar{\gamma}_{i,j} = \sigma_{h_i}^2 P_{i,j}/\sigma_i^2$, with $\sigma_{h_i}^2$ denoting the channel gain variance.

Following the same variable substitution as in P2 and P3, P4 can be equivalently reformulated as a convex problem, from which a semi-closed-form solution is obtained via the KKT conditions below.

Theorem 4 (Semi-closed-form for analog schemes under Rayleigh channels). *The optimal solution of P4 admits the following semi-closed form:*

$$\frac{1}{c_{i,j}N_{i,j}} \sum_{k=1}^{i} 2B_{i,j,k}^* \tau_k = \psi^{-1}\left(\Lambda_i/a_{i,j}\right), \tag{37}$$

$$\frac{P_{i,j}^*}{c_{i,j}N_{i,j}} \sum_{k=1}^{i} 2B_{i,j,k}^* \tau_k = \frac{2\bar{P}Bt_m a_{i,j}\sqrt{G(\psi^{-1}(\Lambda_i/a_{i,j}))}}{\sum_{i=1}^m \sum_{j=1}^{D_i} a_{i,j}c_{i,j}N_{i,j}\sqrt{G(\psi^{-1}(\Lambda_i/a_{i,j}))}}, \tag{38}$$

where $a_{i,j} = \sqrt{\frac{r_i\beta_{i,j}\alpha_{i,j}\sigma_i^2}{\sigma_{h_i}^2 c_{i,j}N_{i,j}}}$, $\phi(T) = \sqrt{G(T)}$, $\psi(T) = -\frac{G'(T)}{2\sqrt{G(T)}}$, $\Lambda_i = \sum_{k\geq i}\mu_k$. Moreover, $\{\mu_k\} \geq 0$ are the optimal dual variables satisfying

$$\mu_k\left(\sum_{i=1}^k \sum_{j=1}^{D_i} c_{i,j}N_{i,j}T_{i,j} - 2Bt_k\right) = 0, \qquad \forall k. \tag{39}$$

Consequently, the overall weighted accuracy degradation is given by

$$\Delta acc = \frac{\left(\sum_{i=1}^m \sum_{j=1}^{D_i} a_{i,j}c_{i,j}N_{i,j}\sqrt{G(\psi^{-1}(\Lambda_i/a_{i,j}))}\right)^2}{2\bar{P}Bt_m \sum_{i=1}^m r_i}. \tag{40}$$

Proof. See Appendix H in technical report [8]. □

Insights: For analog schemes, the Rayleigh fading channel modifies the effective layer importance from $a_{i,j}$ to $a_{i,j}\sqrt{G(\psi^{-1}(\Lambda_i/a_{i,j}))}$, where $G(T)$ quantifies the penalty induced by channel fluctuations. Therefore, unlike the AWGN case, the optimal allocation in Rayleigh fading must account for both task importance and fading uncertainty. Importantly, $G(T)$ approaches 1 as transmissions or bandwidth increase, indicating that retransmissions gradually average out the fading randomness. As a result, the Rayleigh solution asymptotically converges to the AWGN allocation rule.

6.2 Digital Transmissions

Similar to the analog case, our initial step for digital transmission is to characterize the accuracy degradation by evaluating LSNR. By Corollary 1, LSNR is linked to the expected bit error rate (BER) under the threshold-based stopping rule, as given below.

Proposition 4 (Expected BER for Digital Transmission under Rayleigh Channels). *For digital transmission under Rayleigh channels, the expected BER is*

$$P_{b,i,j} = K_{i,j}\left(Q\left(\sqrt{\eta_{i,j}\Gamma_{i,j}}\right) - \sqrt{\frac{\eta_{i,j}\bar{\gamma}_{i,j}}{\eta_{i,j}\bar{\gamma}_{i,j}+2}} e^{\frac{\Gamma_{i,j}}{\bar{\gamma}_{i,j}}} Q\left(\sqrt{(\eta_{i,j}\bar{\gamma}_{i,j}+2)\frac{\Gamma_{i,j}}{\bar{\gamma}_{i,j}}}\right)\right). \tag{41}$$

Proof. The corresponding derivation is provided in Appendix I of technical report [8]. □

By substituting the expected BER expression into our general weighted accuracy-loss framework, the digital resource allocation problem over Rayleigh fading channels is expressed as

$$(P5) \quad \min_{\mathbf{B},\mathbf{P}} \quad \frac{1}{\sum_{i=1}^m r_i} \sum_{i=1}^m r_i \sum_{j=1}^{D_i} \beta_{i,j}\alpha_{i,j}\zeta_{i,j}P_{b,i,j} \quad \text{s.t.} \quad (1)-(5).$$

This problem is non-convex and has a highly involved objective form, which makes it difficult to solve via standard convex optimization techniques. In addition, the variables B and P are intrinsically coupled, further complicating the structure. To obtain an efficient and feasible solution, we develop an iterative algorithm based on alternating optimization (AO) and successive convex approximation (SCA). In particular, we alternately optimize B and P while keeping the other variables fixed. This leads to a sequence of tractable surrogate subproblems that can be solved efficiently. Repeating this procedure yields a suboptimal solution to (P5). The detailed algorithm is summarized in Appendix J of technical report [8].

7 Performance Evaluation

In this section, we conduct performance evaluation to demonstrate the effectiveness of our proposed layer-importance-aware model transmission schemes.

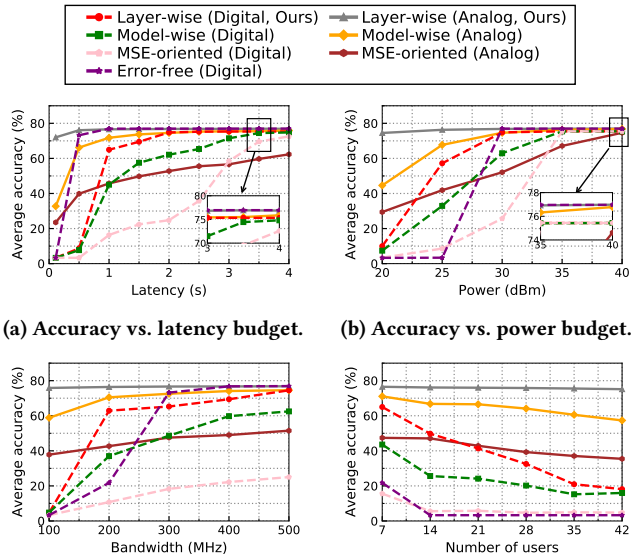
7.1 Experimental Settings

We consider a multi-user AI model downloading system where a BS delivers parameters to multiple UEs over a shared wireless channel. Unless otherwise specified, the total available BS bandwidth is 200 MHz, supporting an effective spatial multiplexing gain of four to serve UEs in parallel. The default system contains 7 UEs randomly distributed within a 1 km^2 area, with equal importance weights assigned to all users. To examine different resource regimes, we vary one system parameter while fixing the others. Specifically, we study the effects of latency under average power budget $\bar{P} = 30$ dBm, the effects of power under latency requirement $t_m = 4$ s, the effects of bandwidth under $t_m = 0.8$ s and $\bar{P} = 30$ dBm, and the effects of the number of users under $t_m = 4$ s and $\bar{P} = 24$ dBm.

Models and Datasets. We consider four representative DNN architectures, namely MobileNet, ShuffleNet, ResNet-18, and VGG-16, which cover both lightweight and relatively large models. These models are assigned to different UEs and evaluated on CIFAR-10, CIFAR-100, and the ImageNet-1K validation set, thereby creating heterogeneous user tasks in terms of model scale, layer structure, and inference difficulty. The reported average accuracy is averaged across all UEs. More detailed model-task assignments and evaluation settings are provided in the technical report [8].

Transmission Setup. For analog transmission, model parameters are directly conveyed by linearly modulated waveforms over the wireless channel. For digital transmission except for the error-free transmission baseline, all model parameters in the digital schemes are uniformly quantized to 6 bits prior to modulation, so as to keep the transmission load practically meaningful under latency constraints. This is because a higher bit-level will significantly increase the transmission delay of digital transmissions with minimal accuracy gain in our settings. Unless otherwise specified, 4-QAM is used for digital transmission, while higher-order modulation is adopted only under stringent latency requirements to reduce the symbol overhead. Both analog and digital schemes are evaluated under the same bandwidth, latency, and average power constraints.

Baselines. We compare our proposed design against three baselines: (1) *Model-wise importance-aware allocation*, which allocates resources based on model-level importance but treats all layers within a model equally, highlighting the value of fine-grained layer-wise scheduling; (2) *MSE-oriented allocation*, a conventional communication baseline minimizing mean-squared reconstruction error rather than maximizing final inference accuracy; and (3) *Error-free transmission*, a digital benchmark that quantizes model parameters to fit the maximum error-free rate under given resource constraints. Specifically, we use finite blocklength theory [21] under a target packet error rate of 10^{-9} to compute the maximum error-free rate, which approximates an error-free transmission setting.



(a) Accuracy vs. latency budget. (b) Accuracy vs. power budget.

(c) Accuracy vs. total bandwidth. (d) Accuracy vs. number of users.

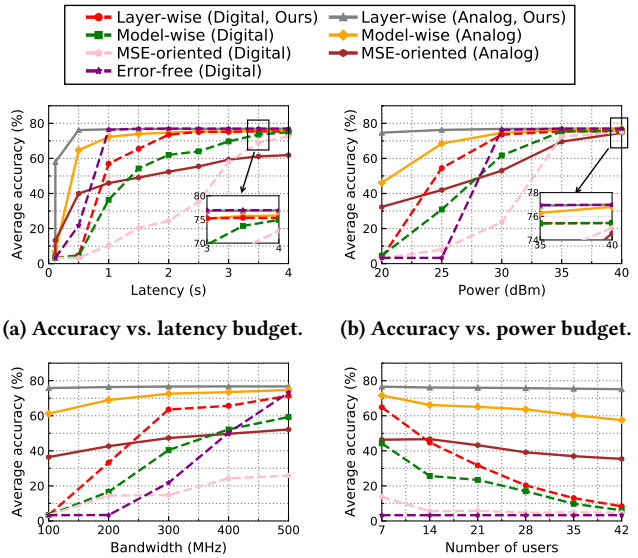
Figure 3: Performance under AWGN channels.

7.2 Performance under AWGN and Rayleigh Channels

Results under AWGN and Rayleigh fading channels are presented in Fig. 3 and Fig. 4, respectively.

Overall Superiority. The proposed layer-importance-aware schemes consistently outperform all baselines across both channel models, which highlights the critical necessity of exploiting fine-grained layer-wise heterogeneity. The *model-wise* baseline, which accounts for macro-level task differences, falls short because it sub-optimally treats all layers within a model equally, thereby wasting valuable resources on less critical parameters. Furthermore, the *MSE-oriented* baseline underperforms because merely minimizing signal reconstruction distortion fails to capture the unequal contributions of different layers to the final inference accuracy. The *error-free* digital benchmark exhibits poor performance, particularly under tight resource constraints. To strictly satisfy the finite blocklength capacity limits without transmission errors, this baseline is forced to quantize model parameters to extremely low bit-widths.

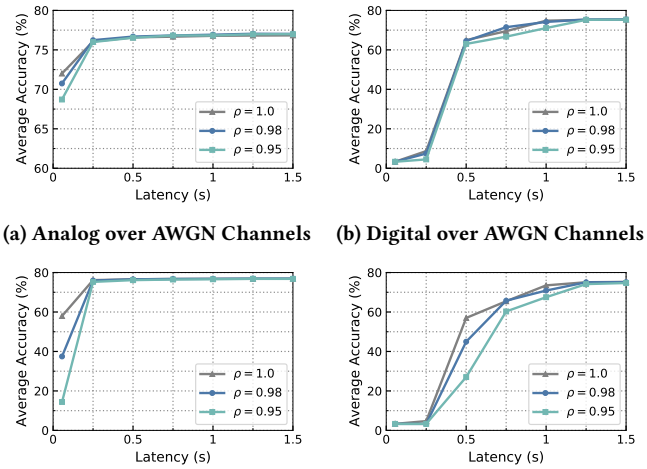
Analog vs. Digital Transmission Paradigms. Analog transmission generally outperforms digital transmission under the same resource budgets, especially in the resource-limited regime. According to Proposition 2, the LSNR in digital transmission is closely related to the BER. When the received SNR decreases, the BER



(a) Accuracy vs. latency budget. (b) Accuracy vs. power budget.

(c) Accuracy vs. total bandwidth. (d) Accuracy vs. number of users.

Figure 4: Performance under Rayleigh channels.



(a) Analog over AWGN Channels (b) Digital over AWGN Channels

(c) Analog over Rayleigh Channels (d) Digital over Rayleigh Channels

Figure 5: Impact of CSI under varied latency budgets.

increases rapidly, which leads to more severe model reconstruction errors and hence larger accuracy degradation. In contrast, analog transmission exhibits a more graceful performance degradation as the channel quality deteriorates.

Latency and Power Efficiency. The proposed analog scheme significantly minimizes the resource overhead required to reach target accuracies. Specifically, Fig. 3a and Fig. 3b illustrate the latency and power trade-offs under AWGN, while their counterparts under Rayleigh fading are depicted in Fig. 4a and Fig. 4b, respectively. Under AWGN with stringent latency constraints, our approach yields a 39.3-percentage-point accuracy gain over the second-best method. Even under the amplified latency sensitivity of Rayleigh fading, the proposed analog scheme successfully maintains strong inference performance while cutting the required latency by 75%. Furthermore, as shown in the low-power regimes of Fig. 3b and Fig. 4b, the proposed method secures a > 30-percentage-point advantage, equivalent to saving approximately 10 dB in power budgets compared to competing schemes.

Bandwidth Efficiency and Scalability. The advantages of our scheme extend to bandwidth utilization and multi-user scalability, as detailed in Fig. 3c and Fig. 3d for AWGN channels, and Fig. 4c and Fig. 4d for Rayleigh channels. The proposed analog scheme reduces the requisite bandwidth by up to 200 MHz to achieve target accuracies. Notably, as the number of users increases and resource contention intensifies, the performance trends in Fig. 3d and Fig. 4d demonstrate that our scheme degrades much more gracefully than baselines, establishing its superior robustness in dense multi-user environments.

Impact of Channel Fading. While similar trends are observed under AWGN and Rayleigh channels, Rayleigh fading causes much more pronounced degradation in resource-limited regimes. This is because deep fading severely limits transmission reliability under tight deadlines, making the performance highly sensitive to latency. For example, near the minimum feasible latency, a further 1% reduction in the latency causes an accuracy drop of over 30%. Conversely, this also highlights the effectiveness of retransmission: even a marginal latency allowance can substantially reduce deep-fading loss. Overall, the proposed analog approach remains robust, reducing the required latency by 75% even under severe fading. Finally, as available resources (latency, power, or bandwidth) increase, fading-induced degradation is gradually mitigated, and the Rayleigh performance approaches the AWGN upper bound.

7.3 Robustness to Imperfect CSI

While perfect CSI is assumed above, practical systems inevitably suffer from channel uncertainty. We evaluate robustness to imperfect CSI using a generic Gauss–Markov formulation [28]:

$$\hat{h} = \rho h + \sqrt{1-\rho^2}\,\epsilon \tag{42}$$

where $\rho \in [0,1]$ denotes the CSI quality (with $\rho = 1$ being perfect CSI) and $\epsilon \sim \mathcal{CN}(0,1)$.

As shown in Fig. 5, our proposed schemes are highly resilient to mild CSI mismatch. When $\rho \geq 0.98$, the performance curves virtually overlap with the perfect CSI baseline, demonstrating negligible accuracy drops across both analog and digital schemes in AWGN environments. The adverse impact of severe CSI imperfection ($\rho = 0.95$) is localized entirely to highly constrained regimes (e.g., extremely low latency under Rayleigh fading), where the compounded effects of deep fading and estimation errors deteriorate the received signal quality. Nonetheless, this minor performance gap vanishes rapidly as latency or power budgets are relaxed, confirming that the proposed layer-importance-aware designs remain highly robust under realistic channel conditions.

8 Conclusion

In this paper, we proposed a layer-importance-aware model transmission framework for multi-user AI model downloading. By analyzing the impact of channel noise on inference accuracy, we established accuracy metrics to characterize the contribution of different model layers. Based on this, the multi-user model transmission problem was formulated as an accuracy degradation minimization problem under latency and average power constraints. For AWGN channels, optimal resource allocation solutions were derived for both analog and digital schemes. For Rayleigh channels, we developed an optimal solution for the analog scheme and an efficient AO-SCA based algorithm for the digital scheme. The results showed that the proposed schemes can significantly improve the inference accuracy while substantially reducing the required communication resources. In particular, the analog layer-wise design demonstrated clear advantages in resource-constrained regimes and exhibited strong robustness under fading channels and imperfect CSI. These results confirm the effectiveness of layer-importance-aware transmission for resource-efficient wireless model delivery.

This paper opens several important directions for future work. First, our analysis relies on a small-perturbation assumption; extending the analytical framework to the high-noise regime, where higher-order terms become non-negligible, would broaden its applicability. Second, extending the framework to multi-cell scenarios, as well as to emerging on-device foundation models, would be valuable for next-generation edge intelligence.

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A Proof of Theorem 1

Under Assumption 1, the locally smoothed performance response can be expanded around f_0 as

$$\mathbb{E}[\Delta A] = \nabla A(f_0)^T \mathbb{E}[\delta f] + \frac{1}{2}\mathbb{E}\left[\delta f^T H \delta f\right] + o\left(\mathbb{E}\left[\|\delta f\|^2\right]\right), \quad (43)$$

where

$$\Delta A \triangleq A(\hat{f}) - A(f_0), \quad (44)$$

$\delta f \triangleq \hat{f} - f_0$, and H denotes the Hessian of $A(\cdot)$ with respect to the logits, evaluated at f_0. This expression provides the link between the logit perturbation and the resulting performance variation, and will be combined with the second-order expansion of $\hat{f}$ with respect to the layer-wise parameter perturbations.

Define the first- and second-order logit perturbation terms as

$$\Delta f^{(1)} \triangleq \sum_{i=1}^{D} J_i \Delta w_i, \qquad \Delta f^{(2)} \triangleq \frac{1}{2}\sum_{i=1}^{D}\sum_{j=1}^{D} \Delta w_i^T G_{i,j} \Delta w_j, \quad (45)$$

such that

$$\hat{f} = f_0 + \Delta f^{(1)} + \Delta f^{(2)} + o(\|\Delta w\|^2). \quad (46)$$

Substituting the above expression into $E[\Delta A]$, and retaining only terms up to second order with respect to Δw, yields

$$\begin{aligned} E[\Delta A] =& \nabla A(f_0)^T E[\Delta f^{(1)}] + \nabla A(f_0)^T E[\Delta f^{(2)}] \\ &+ \frac{1}{2} E\left[(\Delta f^{(1)})^T H \Delta f^{(1)}\right] + o(\|\Delta w\|^2). \end{aligned} \quad (47)$$

Here, the terms involving $(\Delta f^{(2)})^T H \Delta f^{(2)}$ and the cross products between $\Delta f^{(1)}$ and $\Delta f^{(2)}$ are of order higher than two and are therefore omitted.

We first evaluate the linear term. By Assumption 3, $E[\Delta w_i] = 0$ for all i, and thus

$$E[\Delta f^{(1)}] = \sum_{i=1}^{D} J_i E[\Delta w_i] = 0. \quad (48)$$

Therefore, the first term in (47) vanishes.

Next, consider $E[\Delta f^{(2)}]$. For the k-th logit component,

$$[\Delta f^{(2)}]_k = \frac{1}{2}\sum_{i=1}^{D}\sum_{j=1}^{D} \Delta w_i^T G_{i,j}^{(k)} \Delta w_j, \quad (49)$$

where $G_{i,j}^{(k)}$ denotes the (i,j)-th second-order derivative block of the k-th logit component. Taking expectation gives

$$E\left[[\Delta f^{(2)}]_k\right] = \frac{1}{2}\sum_{i=1}^{D}\sum_{j=1}^{D} E\left[\Delta w_i^T G_{i,j}^{(k)} \Delta w_j\right]. \quad (50)$$

For $i \neq j$, Assumption 3 implies that Δw_i and Δw_j are independent and zero-mean, hence

$$E\left[\Delta w_i^T G_{i,j}^{(k)} \Delta w_j\right] = E[\Delta w_i]^T G_{i,j}^{(k)} E[\Delta w_j] = 0. \quad (51)$$

Therefore, only the self-interaction terms remain:

$$E\left[[\Delta f^{(2)}]_k\right] = \frac{1}{2}\sum_{i=1}^{D} E\left[\Delta w_i^T G_{i,i}^{(k)} \Delta w_i\right]. \quad (52)$$

It follows that

$$\nabla A(f_0)^T E[\Delta f^{(2)}] = \frac{1}{2}\sum_{k=1}^{F} \nabla A_{(k)} \sum_{i=1}^{D} E\left[\Delta w_i^T G_{i,i}^{(k)} \Delta w_i\right]. \quad (53)$$

Now consider the quadratic term in (47):

$$\begin{aligned} E\left[(\Delta f^{(1)})^T H \Delta f^{(1)}\right] &= E\left[\left(\sum_{i=1}^{D} J_i \Delta w_i\right)^T H \left(\sum_{j=1}^{D} J_j \Delta w_j\right)\right] \\ &= \sum_{i=1}^{D}\sum_{j=1}^{D} E\left[\Delta w_i^T J_i^T H J_j \Delta w_j\right]. \end{aligned} \quad (54)$$

Again, for $i \neq j$, independence and zero-mean imply

$$E\left[\Delta w_i^T J_i^T H J_j \Delta w_j\right] = E[\Delta w_i]^T J_i^T H J_j E[\Delta w_j] = 0. \quad (55)$$

Thus,

$$E\left[(\Delta f^{(1)})^T H \Delta f^{(1)}\right] = \sum_{i=1}^{D} E\left[\Delta w_i^T J_i^T H J_i \Delta w_i\right]. \quad (56)$$

Substituting (53) and (56) into (47), we obtain

$$\begin{aligned} E[\Delta A] &= \frac{1}{2}\sum_{k=1}^{F} \nabla A_{(k)} \sum_{i=1}^{D} E\left[\Delta w_i^T G_{i,i}^{(k)} \Delta w_i\right] \\ &\quad + \frac{1}{2}\sum_{i=1}^{D} E\left[\Delta w_i^T J_i^T H J_i \Delta w_i\right] + o(\|\Delta w\|^2) \\ &= \frac{1}{2}\sum_{i=1}^{D} E\left[\Delta w_i^T \left(\sum_{k=1}^{F} \nabla A_{(k)} G_{i,i}^{(k)} + J_i^T H J_i\right) \Delta w_i\right] + o(\|\Delta w\|^2). \end{aligned} \quad (57)$$

Using the definition of M_i in (10), we have

$$E[\Delta A] = \frac{1}{2}\sum_{i=1}^{D} E\left[\Delta w_i^T (M_i + J_i^T H J_i) \Delta w_i\right] + o(\|\Delta w\|^2). \quad (58)$$

Finally, applying the identity

$$E[x^T A x] = \mathrm{tr}(A\, E[x x^T]), \quad (59)$$

for any random vector x and constant matrix A of compatible dimensions, together with $\Sigma_i = E[\Delta w_i \Delta w_i^T]$, yields

$$E[\Delta A] = \frac{1}{2}\sum_{i=1}^{D} \mathrm{tr}\left((M_i + J_i^T H J_i)\Sigma_i\right) + o(\|\Delta w\|^2). \quad (60)$$

Neglecting the higher-order remainder term under the small-perturbation assumption completes the proof.

B Proof of Proposition 1

Consider a parameter w transmitted through analog modulation. Let the received signal after retransmission and combining be written as

$$y = w + n_{\mathrm{eq}}, \quad (61)$$

where n_{eq} denotes the equivalent additive noise after combining. Then the reconstruction error is

$$e_w = \hat{w} - w = n_{\mathrm{eq}}. \quad (62)$$

For a given realization of the effective combined SNR γ^{tot}, the conditional distortion power satisfies

$$\mathbb{E}\left[e_w^2 \mid \gamma^{\text{tot}}\right] = \frac{\mathbb{E}[w^2]}{\gamma^{\text{tot}}}, \tag{63}$$

since the received SNR is defined as the ratio between the signal power and the equivalent noise power after combining. Taking expectation over γ^{tot}, we have

$$\begin{aligned}\mathbb{E}[e_w^2] &= \mathbb{E}_{\gamma^{\text{tot}}}\left[\mathbb{E}\left[e_w^2 \mid \gamma^{\text{tot}}\right]\right] \\ &= \mathbb{E}_{\gamma^{\text{tot}}}\left[\frac{\mathbb{E}[w^2]}{\gamma^{\text{tot}}}\right] \\ &= \mathbb{E}[w^2]\,\mathbb{E}\left[\frac{1}{\gamma^{\text{tot}}}\right].\end{aligned} \tag{64}$$

Applying the same relation to all parameters in layer (i, j), the normalized layer distortion becomes

$$\begin{aligned}\frac{1}{LSNR_{i,j}} &= \frac{\mathbb{E}\left[\|\Delta \mathbf{w}_{i,j}\|^2\right]}{\mathbb{E}\left[\|\mathbf{w}_{i,j}\|^2\right]} \\ &= \mathbb{E}\left[\frac{1}{\gamma_{i,j}^{\text{tot}}}\right].\end{aligned} \tag{65}$$

This completes the proof.

C Proof of Proposition 2

For the j-th layer of the i-th UE, by the LSNR definition, we have

$$\frac{1}{LSNR_{i,j}} = \frac{\mathbb{E}\left[\|\Delta \mathbf{w}_{i,j}\|^2\right]}{\mathbb{E}\left[\|\mathbf{w}_{i,j}\|^2\right]}. \tag{66}$$

Furthermore, define $s_k \in \{0, 1\}$ as the error indicator of the k-th bit, where $s_k = 1$ indicates that the bit is flipped during transmission and $s_k = 0$ otherwise. Then the reconstruction error of this parameter can be expressed as

$$e_w = \hat{w} - w = \Delta \sum_{k=0}^{b-1} s_k (1 - 2n_k) 2^k. \tag{67}$$

Substituting (67) into the numerator gives

$$\begin{aligned}\mathbb{E}\left[\|\Delta \mathbf{w}_{i,j}\|^2\right] &= \mathbb{E}\left[\sum_{\ell=1}^{N_{i,j}} \left(\Delta \sum_{k=0}^{b-1} s_{\ell,k}(1 - 2n_{\ell,k}) 2^k\right)^2\right] \\ &= \Delta^2 \mathbb{E}\left[\sum_{\ell=1}^{N_{i,j}} \left(\sum_{k=0}^{b-1} s_{\ell,k}(1 - 2n_{\ell,k}) 2^k\right)^2\right],\end{aligned} \tag{68}$$

where $n_{\ell,k}$ and $s_{\ell,k}$ denote the original bit and the error indicator of the k-th bit of the ℓ-th parameter, respectively.

Expanding the square term yields

$$\begin{aligned}\left(\sum_{k=0}^{b-1} s_{\ell,k}(1 - 2n_{\ell,k}) 2^k\right)^2 &= \sum_{k=0}^{b-1} s_{\ell,k}^2 (1 - 2n_{\ell,k})^2 4^k \\ &\quad + \sum_{k \neq t} s_{\ell,k} s_{\ell,t} (1 - 2n_{\ell,k})(1 - 2n_{\ell,t}) 2^{k+t}.\end{aligned} \tag{69}$$

Since $s_{\ell,k} \in \{0, 1\}$, we have $s_{\ell,k}^2 = s_{\ell,k}$, and since $n_{\ell,k} \in \{0, 1\}$, we have $(1 - 2n_{\ell,k})^2 = 1$. Therefore, the first term becomes

$$\sum_{k=0}^{b-1} s_{\ell,k} 4^k. \tag{70}$$

Under the assumption that bit flips are independent and identically distributed with probability $P_{b,i,j}$, we have

$$\mathbb{E}[s_{\ell,k}] = P_{b,i,j}, \tag{71}$$

and thus

$$\mathbb{E}\left[\sum_{k=0}^{b-1} s_{\ell,k} 4^k\right] = P_{b,i,j} \sum_{k=0}^{b-1} 4^k. \tag{72}$$

For the cross terms with $k \neq t$, their expectation is proportional to

$$\mathbb{E}[s_{\ell,k} s_{\ell,t}] = P_{b,i,j}^2. \tag{73}$$

Hence, these terms are of order $O(P_{b,i,j}^2)$. In the practical operating regime where $P_{b,i,j} \ll 1$, such higher-order terms are negligible compared with the linear term in $P_{b,i,j}$. Therefore,

$$\mathbb{E}\left[\|\Delta \mathbf{w}_{i,j}\|^2\right] \approx N_{i,j} \Delta^2 P_{b,i,j} \sum_{k=0}^{b-1} 4^k. \tag{74}$$

Substituting (74) into (66), we obtain

$$\begin{aligned}\frac{1}{LSNR_{i,j}} &\approx \frac{N_{i,j} \Delta^2 P_{b,i,j} \sum_{k=0}^{b-1} 4^k}{\mathbb{E}\left[\|\mathbf{w}_{i,j}\|^2\right]} \\ &= \underbrace{\frac{N_{i,j} \Delta^2 \sum_{k=0}^{b-1} 4^k}{\mathbb{E}\left[\|\mathbf{w}_{i,j}\|^2\right]}}_{\zeta_{i,j}} P_{b,i,j}.\end{aligned} \tag{75}$$

This proves (22).

D Empirical Estimation of α_i and β_i

The estimation procedure of α_i and β_i is outlined in Algorithm 1.

E Proof of Theorem 2

Introduce $z_{i,j} = T_{i,j} P_{i,j}$, the analog transmission problem can be equivalently rewritten as

$$\min_{\mathbf{z}} \quad \frac{1}{\sum_{i=1}^m r_i} \sum_{i=1}^m \sum_{j=1}^{D_i} \frac{r_i \beta_{i,j} \alpha_{i,j} \sigma_i^2}{h_i^2 z_{i,j}} \tag{76}$$

subject to

$$\sum_{i=1}^m \sum_{j=1}^{D_i} c_{i,j} N_{i,j} z_{i,j} \leq \bar{P} C_m, \qquad z_{i,j} > 0, \ \forall i, j, \tag{77}$$

where C_m denotes the total symbol budget, i.e., the maximum number of symbols that can be transmitted when the available bandwidth is fully utilized. Since the bandwidth allocation variables $T_{i,j}$ do not appear explicitly in either the objective function or the power constraint after the substitution, the bandwidth-related constraints in (3) and (2) can be handled separately after the optimal closed-form solution of $\mathbf{z}$ is obtained.

Since $1/z_{i,j}$ is convex over $z_{i,j} > 0$, problem (76)–(77) is convex. Moreover, the objective function is strictly decreasing with respect

Algorithm 1 Empirical Estimation of α_i and β_i

Require: Model $f(\cdot;\mathbf{w})$, validation set $\mathcal{D}_{\text{val}}$, samples N_{s}, prescribed SNR_i, thresholds $\Delta Acc_{\min}$ and $\Delta Acc_{\max}$
Ensure: $\{\alpha_i, \beta_i\}_{i=1}^{D}$

Compute the original logits $f_0(x)$ and accuracy Acc_0 on $\mathcal{D}_{\text{val}}$
for each layer $i = 1, \ldots, D$ **do**
  Set $S_\alpha = 0$, $S_\beta = 0$, and $C_\beta = 0$
  for $n = 1, \ldots, N_{\text{s}}$ **do**
    Inject random noise into layer i according to SNR_i
    Compute logits $\hat{f}_i^{(n)}(x)$ and accuracy $\widehat{Acc}_i^{(n)}$
    Compute
$$D_i^{(n)} = \frac{1}{|\mathcal{D}_{\text{val}}|} \sum_{x \in \mathcal{D}_{\text{val}}} \left\| \hat{f}_i^{(n)}(x) - f_0(x) \right\|_2^2$$
    Update $S_\alpha \leftarrow S_\alpha + D_i^{(n)} LSNR_i$
    Compute $\Delta Acc_i^{(n)} = Acc_0 - \widehat{Acc}_i^{(n)}$
    if $\Delta Acc_{\min} \le \Delta Acc_i^{(n)} \le \Delta Acc_{\max}$ **then**
      Update $S_\beta \leftarrow S_\beta + \Delta Acc_i^{(n)} / D_i^{(n)}$
      Update $C_\beta \leftarrow C_\beta + 1$
    end if
  end for
  Set $\alpha_i = S_\alpha / N_{\text{s}}$
  Set $\beta_i = S_\beta / C_\beta$
end for
return $\{\alpha_i, \beta_i\}_{i=1}^{D}$

to each $z_{i,j}$. Therefore, the power constraint in (77) must be active at the optimum, i.e.,

$$\sum_{i=1}^{m} \sum_{j=1}^{D_i} c_{i,j} N_{i,j} z_{i,j} = \bar{P} C_m. \tag{78}$$

For convenience, define

$$a_{i,j} \triangleq \sqrt{\frac{r_i \beta_{i,j} \alpha_{i,j} \sigma_i^2}{h_i^2 c_{i,j} N_{i,j}}}. \tag{79}$$

The Lagrangian of the problem is

$$\mathcal{L}(\mathbf{z}, \lambda) = \frac{1}{\sum_{i=1}^{m} r_i} \sum_{i=1}^{m} \sum_{j=1}^{D_i} \frac{r_i \beta_{i,j} \alpha_{i,j} \sigma_i^2}{h_i^2 z_{i,j}} + \lambda \left(\sum_{i=1}^{m} \sum_{j=1}^{D_i} c_{i,j} N_{i,j} z_{i,j} - \bar{P} C_m \right). \tag{80}$$

From the stationarity condition, we have

$$-\frac{r_i \beta_{i,j} \alpha_{i,j} \sigma_i^2}{h_i^2 z_{i,j}^2 \sum_{i=1}^{m} r_i} + \lambda c_{i,j} N_{i,j} = 0, \tag{81}$$

which gives

$$z_{i,j} = \frac{a_{i,j}}{\sqrt{\lambda \sum_{i=1}^{m} r_i}}. \tag{82}$$

Substituting (82) into (78) yields

$$\frac{1}{\sqrt{\lambda \sum_{i=1}^{m} r_i}} = \frac{\bar{P} C_m}{\sum_{i=1}^{m} \sum_{j=1}^{D_i} c_{i,j} N_{i,j} a_{i,j}}. \tag{83}$$

Hence, the optimal solution is

$$z_{i,j}^* = \frac{\bar{P} C_m \, a_{i,j}}{\sum_{i=1}^{m} \sum_{j=1}^{D_i} c_{i,j} N_{i,j} a_{i,j}}, \tag{84}$$

which proves (27).

Substituting $z_{i,j}^*$ into the weighted accuracy-loss expression gives

$$\Delta acc_i = \frac{1}{r_i} \sum_{j=1}^{D_i} \frac{\beta_{i,j} \alpha_{i,j} \sigma_i^2}{h_i^2 z_{i,j}^*} \tag{85}$$

$$= \frac{1}{r_i} \sum_{j=1}^{D_i} \frac{\beta_{i,j} \alpha_{i,j} \sigma_i^2}{h_i^2} \cdot \frac{\sum_{u=1}^{m} \sum_{v=1}^{D_u} c_{u,v} N_{u,v} a_{u,v}}{\bar{P} C_m \, a_{i,j}} \tag{86}$$

$$= \frac{\sum_{u=1}^{m} \sum_{v=1}^{D_u} c_{u,v} N_{u,v} a_{u,v}}{\bar{P} C_m \, r_i} \sum_{j=1}^{D_i} c_{i,j} N_{i,j} a_{i,j}, \tag{87}$$

which proves (29).

Therefore, the overall weighted accuracy loss is

$$\Delta acc = \frac{1}{\sum_{i=1}^{m} r_i} \sum_{i=1}^{m} r_i \Delta acc_i \tag{88}$$

$$= \frac{1}{\sum_{i=1}^{m} r_i} \sum_{i=1}^{m} \frac{\sum_{u=1}^{m} \sum_{v=1}^{D_u} c_{u,v} N_{u,v} a_{u,v}}{\bar{P} C_m} \sum_{j=1}^{D_i} c_{i,j} N_{i,j} a_{i,j} \tag{89}$$

$$= \frac{\left(\sum_{i=1}^{m} \sum_{j=1}^{D_i} c_{i,j} N_{i,j} a_{i,j} \right)^2}{\bar{P} C_m \sum_{i=1}^{m} r_i}, \tag{90}$$

which proves (30). The proof is thus completed.

F Proof of Theorem 3

Introduce $z_{i,j} = T_{i,j} P_{i,j}$, the digital transmission problem can be equivalently rewritten as

$$\min_{\mathbf{z}} \quad \frac{1}{\sum_{i=1}^{m} r_i} \sum_{i=1}^{m} \sum_{j=1}^{D_i} d_{i,j} Q\left(\sqrt{e_{i,j} z_{i,j}}\right) \tag{91}$$

subject to

$$\sum_{i=1}^{m} \sum_{j=1}^{D_i} c_{i,j} N_{i,j} z_{i,j} \le \bar{P} C_m, \qquad z_{i,j} > 0, \ \forall i, j, \tag{92}$$

where

$$d_{i,j} = \frac{r_i \beta_{i,j} \alpha_{i,j} \zeta_{i,j} K_{i,j}}{c_{i,j} N_{i,j}}, \qquad e_{i,j} = \frac{\eta_{i,j} h_i^2}{\sigma_i^2}. \tag{93}$$

Since the bandwidth allocation variables $T_{i,j}$ do not appear explicitly in either the objective function or the power constraint after the substitution, the bandwidth-related constraints in (3) and (2) can be handled separately after the optimal semi-closed-form solution of $\mathbf{z}$ is obtained.

To establish convexity, define

$$f_{i,j}(z) = Q\left(\sqrt{e_{i,j} z}\right), \qquad z > 0.$$

Its first derivative is

$$f'_{i,j}(z) = -\frac{\sqrt{e_{i,j}}}{2\sqrt{2\pi z}} \exp\left(-\frac{e_{i,j} z}{2}\right), \tag{94}$$

and its second derivative is

$$f''_{i,j}(z) = \frac{\sqrt{e_{i,j}}}{4\sqrt{2\pi}} \exp\left(-\frac{e_{i,j} z}{2}\right) \left(z^{-3/2} + e_{i,j} z^{-1/2}\right) > 0. \tag{95}$$

Therefore, $f_{i,j}(z)$ is convex for $z > 0$, and hence problem (91)–(92) is convex. Moreover, the objective function is strictly decreasing with respect to each $z_{i,j}$. Therefore, the power constraint in (92) must be active at the optimum, i.e.,

$$\sum_{i=1}^{m}\sum_{j=1}^{D_i} c_{i,j}N_{i,j}z_{i,j} = \bar{P}\,C_m. \tag{96}$$

The Lagrangian of the problem is

$$\mathcal{L}(\mathbf{z},\lambda) = \frac{1}{\sum_{i=1}^{m} r_i}\sum_{i=1}^{m}\sum_{j=1}^{D_i} d_{i,j}Q\big(\sqrt{e_{i,j}z_{i,j}}\big)+\lambda\left(\sum_{i=1}^{m}\sum_{j=1}^{D_i} c_{i,j}N_{i,j}z_{i,j} - \bar{P}C_m\right). \tag{97}$$

From the stationarity condition, we have

$$-\frac{d_{i,j}\sqrt{e_{i,j}}}{2\sqrt{2\pi z_{i,j}}\sum_{i=1}^{m} r_i}\exp\Big(-\frac{e_{i,j}z_{i,j}}{2}\Big) + \lambda c_{i,j}N_{i,j} = 0. \tag{98}$$

Equivalently,

$$\frac{d_{i,j}\sqrt{e_{i,j}}}{2\sqrt{2\pi z_{i,j}}\sum_{i=1}^{m} r_i}\exp\Big(-\frac{e_{i,j}z_{i,j}}{2}\Big) = \lambda c_{i,j}N_{i,j}. \tag{99}$$

Multiplying both sides of (99) by $\sum_{i=1}^{m} r_i$ and absorbing the constant factor into the Lagrange multiplier, (99) can be rewritten as

$$\frac{d_{i,j}\sqrt{e_{i,j}}}{2\sqrt{2\pi z_{i,j}}}\exp\Big(-\frac{e_{i,j}z_{i,j}}{2}\Big) = \lambda c_{i,j}N_{i,j}. \tag{100}$$

Squaring both sides of (100) and rearranging terms yield

$$(e_{i,j}z_{i,j})\exp(e_{i,j}z_{i,j}) = \frac{d_{i,j}^2 e_{i,j}^2}{8\pi\lambda^2 c_{i,j}^2 N_{i,j}^2}. \tag{101}$$

Thus, by the definition of the Lambert-W function,

$$z_{i,j}^* = \frac{1}{e_{i,j}}W\left(\frac{d_{i,j}^2 e_{i,j}^2}{8\pi\lambda^2 c_{i,j}^2 N_{i,j}^2}\right). \tag{102}$$

By further absorbing $c_{i,j}N_{i,j}$ into the multiplier, (102) can be equivalently expressed as

$$z_{i,j}^* = \frac{1}{e_{i,j}}W\left(\frac{d_{i,j}^2 e_{i,j}^2}{8\pi(\lambda^*)^2}\right), \tag{103}$$

which proves (35).

Substituting $z_{i,j}^*$ into the active constraint (96) gives

$$\sum_{i=1}^{m}\sum_{j=1}^{D_i}\frac{c_{i,j}N_{i,j}}{e_{i,j}}W\left(\frac{d_{i,j}^2 e_{i,j}^2}{8\pi(\lambda^*)^2}\right) = \bar{P}\,C_m. \tag{104}$$

The scalar λ^* can be efficiently obtained by solving the resulting one-dimensional monotone equation via bisection. The proof is thus completed.

G Proof of Proposition 3

Under Rayleigh fading, the instantaneous received SNR is exponentially distributed as

$$\gamma \sim \mathrm{Exp}(\bar{\gamma}),$$

where $\bar{\gamma}$ denotes the average received SNR per transmission. After k transmissions with MRC, the combined SNR s_k follows a Gamma distribution with PDF

$$f_{s,k}(s) = \frac{s^{k-1}}{\bar{\gamma}^k(k-1)!}e^{-s/\bar{\gamma}}, \qquad s \geq 0. \tag{105}$$

Let Γ denote the stopping threshold. The event $T = k$ means that the accumulated SNR is below Γ after $k-1$ transmissions and reaches Γ at the k-th transmission. Hence, for $s_0 \geq \Gamma$,

$$\begin{aligned} g_{s,T}(s = s_0, T = k) &= \int_0^{\Gamma} f_{s,k-1}(s)f_{s,1}(s_0 - s)\,ds \\ &= \int_0^{\Gamma}\frac{s^{k-2}}{\bar{\gamma}^{k-1}(k-2)!}e^{-s/\bar{\gamma}}\cdot\frac{1}{\bar{\gamma}}e^{-(s_0-s)/\bar{\gamma}}\,ds \\ &= \frac{\Gamma^{k-1}}{\bar{\gamma}^k(k-1)!}e^{-s_0/\bar{\gamma}}. \end{aligned} \tag{106}$$

Therefore, the PDF of the stopping SNR is

$$\begin{aligned} f_s(s) &= \sum_{k=1}^{\infty} g_{s,T}(s = s, T = k) \\ &= \sum_{k=1}^{\infty}\frac{\Gamma^{k-1}}{\bar{\gamma}^k(k-1)!}e^{-s/\bar{\gamma}} \\ &= \frac{1}{\bar{\gamma}}e^{-s/\bar{\gamma}}\sum_{k=1}^{\infty}\frac{(\Gamma/\bar{\gamma})^{k-1}}{(k-1)!} \\ &= \frac{1}{\bar{\gamma}}e^{-(s-\Gamma)/\bar{\gamma}}, \qquad s \geq \Gamma, \end{aligned} \tag{107}$$

which is a shifted exponential distribution.

The stopping probability at the k-th transmission is

$$\begin{aligned} \Pr(T = k) &= \int_{\Gamma}^{\infty} g_{s,T}(s = s, T = k)\,ds \\ &= \frac{(\Gamma/\bar{\gamma})^{k-1}}{(k-1)!}e^{-\Gamma/\bar{\gamma}}. \end{aligned} \tag{108}$$

Hence, $T - 1 \sim \mathrm{Poisson}(\Gamma/\bar{\gamma})$, and

$$\mathbb{E}[T] = 1 + \frac{\Gamma}{\bar{\gamma}}. \tag{109}$$

For the analog case,

$$\begin{aligned} \frac{1}{LSNR} &= \mathbb{E}\left[\frac{1}{\gamma_{\mathrm{tot}}}\right] \\ &= \int_{\Gamma}^{\infty}\frac{1}{\gamma_{\mathrm{tot}}}\frac{1}{\bar{\gamma}}e^{-(\gamma_{\mathrm{tot}}-\Gamma)/\bar{\gamma}}\,d\gamma_{\mathrm{tot}} \\ &= \frac{1}{\bar{\gamma}}e^{\Gamma/\bar{\gamma}}\int_{\Gamma}^{\infty}\frac{1}{\gamma_{\mathrm{tot}}}e^{-\gamma_{\mathrm{tot}}/\bar{\gamma}}\,d\gamma_{\mathrm{tot}} \\ &= \frac{1}{\bar{\gamma}}e^{\Gamma/\bar{\gamma}}E_1\left(\frac{\Gamma}{\bar{\gamma}}\right), \end{aligned} \tag{110}$$

where $E_1(x) = \int_x^{\infty}\frac{e^{-t}}{t}\,dt$ is the exponential integral.

H Proof of Theorem 4

H.1 Convexity of $\sqrt{G(x)}$

Lemma 1. *For*

$$G(x) = xe^{x-1}E_1(x-1), \qquad x > 1,$$

the function $\sqrt{G(x)}$ is strictly convex on $(1,\infty)$.

Proof. Let

$$t = x - 1 > 0, \qquad y(t) = e^t E_1(t).$$

Then

$$G(x) = xe^{x-1}E_1(x-1) = (t+1)y(t),$$

and hence

$$\sqrt{G(x)} = \sqrt{(t+1)y(t)}.$$

Since $dt/dx = 1$, it suffices to prove that $\sqrt{(t+1)y(t)}$ is strictly convex for $t > 0$.

Using

$$y'(t) = y(t) - \frac{1}{t},$$

we obtain

$$\frac{d^2}{dt^2}\sqrt{(t+1)y(t)} = \frac{\Phi(t,y(t))}{4t^2(t+1)^{3/2}y(t)^{3/2}},$$

where

$$\Phi(t,y) = t^2(t^2+4t+2)y^2 + 2(t+1)y - (t+1)^2.$$

Since $t > 0$ and $y(t) > 0$, the denominator is positive. Thus it remains to show

$$\Phi(t,y(t)) > 0.$$

Moreover, $\Phi(t,y)$ is increasing in y, because

$$\frac{\partial\Phi(t,y)}{\partial y} = 2t^2(t^2+4t+2)y + 2(t+1) > 0.$$

From the continued-fraction expansion of $e^tE_1(t)$, we have the lower bound

$$y(t) = e^tE_1(t) \geq \frac{t+3}{t^2+4t+2}, \qquad t > 0.$$

Therefore,

$$\Phi(t,y(t)) \geq \Phi\left(t, \frac{t+3}{t^2+4t+2}\right) = \frac{4}{t^2+4t+2} > 0.$$

Hence

$$\frac{d^2}{dt^2}\sqrt{(t+1)y(t)} > 0, \qquad t > 0,$$

and equivalently

$$\frac{d^2}{dx^2}\sqrt{G(x)} > 0, \qquad x > 1.$$

Thus, $\sqrt{G(x)}$ is strictly convex on $(1,\infty)$. □

H.2 Semi-closed-form for P4

Introduce $z_{i,j} = T_{i,j}P_{i,j}$, then P4 can be equivalently rewritten as

$$\min_{\mathbf{T},\mathbf{z}} \quad \frac{1}{\sum_{i=1}^m r_i}\sum_{i=1}^m\sum_{j=1}^{D_i}\frac{r_i\beta_{i,j}\alpha_{i,j}\sigma_i^2}{h_i^2}\frac{G(T_{i,j})}{z_{i,j}} \tag{111}$$

subject to

$$\sum_{i=1}^k\sum_{j=1}^{D_i} c_{i,j}N_{i,j}T_{i,j} \leq C_k, \qquad \forall k, \tag{112}$$

$$\sum_{i=1}^m\sum_{j=1}^{D_i} c_{i,j}N_{i,j}z_{i,j} \leq \bar{P}C_m, \tag{113}$$

where (113) follows from the fact that the last deadline constraint is active at optimum.

For given $\mathbf{T}$, the subproblem with respect to $\mathbf{z}$ is convex. Its Lagrangian is

$$\mathcal{L}_1 = \sum_{i=1}^m\sum_{j=1}^{D_i}\frac{r_i\beta_{i,j}\alpha_{i,j}\sigma_i^2}{h_i^2}\frac{G(T_{i,j})}{z_{i,j}} + \lambda\left(\sum_{i=1}^m\sum_{j=1}^{D_i} c_{i,j}N_{i,j}z_{i,j} - \bar{P}C_m\right), \tag{114}$$

where $\lambda \geq 0$. From the KKT stationarity condition,

$$-\frac{r_i\beta_{i,j}\alpha_{i,j}\sigma_i^2}{h_i^2}\frac{G(T_{i,j})}{z_{i,j}^2} + \lambda c_{i,j}N_{i,j} = 0, \tag{115}$$

which yields

$$z_{i,j}^* = \sqrt{\frac{r_i\beta_{i,j}\alpha_{i,j}\sigma_i^2\,G(T_{i,j})}{\lambda h_i^2 c_{i,j}N_{i,j}}}. \tag{116}$$

Since (113) is active at optimum, λ is determined by

$$\sum_{i=1}^m\sum_{j=1}^{D_i} c_{i,j}N_{i,j}z_{i,j}^* = \bar{P}C_m, \tag{117}$$

and thus

$$z_{i,j}^* = \frac{\bar{P}C_m\sqrt{\dfrac{r_i\beta_{i,j}\alpha_{i,j}\sigma_i^2\,G(T_{i,j})}{h_i^2c_{i,j}N_{i,j}}}}{\sum\limits_{u=1}^m\sum\limits_{v=1}^{D_u}\sqrt{\dfrac{r_u\beta_{u,v}\alpha_{u,v}\sigma_u^2}{h_u^2}\,c_{u,v}N_{u,v}G(T_{u,v})}}. \tag{118}$$

Define

$$a_{i,j} \triangleq \sqrt{\frac{r_i\beta_{i,j}\alpha_{i,j}\sigma_i^2}{h_i^2c_{i,j}N_{i,j}}}. \tag{119}$$

Then (118) can be rewritten as

$$z_{i,j}^* = \frac{\bar{P}C_m\,a_{i,j}\sqrt{G(T_{i,j})}}{\sum\limits_{u=1}^m\sum\limits_{v=1}^{D_u} a_{u,v}c_{u,v}N_{u,v}\sqrt{G(T_{u,v})}}. \tag{120}$$

Substituting (120) into (111), the original problem reduces to

$$\min_{\mathbf{T}} \quad \sum_{i=1}^m\sum_{j=1}^{D_i} a_{i,j}c_{i,j}N_{i,j}\phi(T_{i,j}) \tag{121}$$

subject to (112), where

$$\phi(T) \triangleq \sqrt{G(T)}. \tag{122}$$

Since $\phi(T) = \sqrt{G(T)}$ is convex, thus (121) is a convex optimization problem.

Let $\mu_k \geq 0$ denote the dual variable associated with the k-th deadline constraint in (112). The corresponding Lagrangian is

$$\mathcal{L}_2 = \sum_{i=1}^m\sum_{j=1}^{D_i} a_{i,j}c_{i,j}N_{i,j}\phi(T_{i,j}) + \sum_{k=1}^m\mu_k\left(\sum_{i=1}^k\sum_{j=1}^{D_i} c_{i,j}N_{i,j}T_{i,j} - C_k\right). \tag{123}$$

Since $T_{i,j}$ appears in all cumulative constraints with indices $k \geq i$, the stationarity condition gives

$$a_{i,j}\phi'(T_{i,j}^*) + \sum_{k\geq i}\mu_k = 0. \tag{124}$$

Define

$$\Lambda_i \triangleq \sum_{k\geq i}\mu_k, \qquad \psi(T) \triangleq -\phi'(T) = -\frac{G'(T)}{2\sqrt{G(T)}}. \tag{125}$$

Then

$$\psi(T_{i,j}^*) = \frac{\Lambda_i}{a_{i,j}}, \tag{126}$$

which leads to

$$T_{i,j}^* = \psi^{-1}\left(\frac{\Lambda_i}{a_{i,j}}\right). \tag{127}$$

Together with (120), this proves the semi-closed-form solution.

Finally, substituting (127) into the reduced objective yields

$$\Delta acc = \frac{\left(\sum_{i=1}^{m}\sum_{j=1}^{D_i} a_{i,j}c_{i,j}N_{i,j}\sqrt{G(T_{i,j}^*)}\right)^2}{\bar{P}C_m\sum_{i=1}^{m} r_i}. \tag{128}$$

The semi-closed-form solution can be efficiently computed via an active-set search over the cumulative deadline constraints. Specifically, starting from the last deadline constraint, the active set is progressively updated by adding violated constraints and removing those with negative multipliers, until the KKT conditions are satisfied. The proof is completed.

I Proof of Proposition 4

For the digital transmission scheme, the instantaneous BER is approximated by

$$\hat{P}_b = KQ\left(\sqrt{\eta\gamma_{\text{tot}}}\right), \tag{129}$$

where K and η are modulation-related constants. Under the threshold-based SNR control strategy, the stopping SNR γ_{tot} follows the shifted exponential distribution

$$f_{\gamma_{\text{tot}}}(\gamma) = \frac{1}{\bar{\gamma}}\exp\left(-\frac{\gamma-\Gamma}{\bar{\gamma}}\right), \qquad \gamma \geq \Gamma, \tag{130}$$

where Γ is the stopping threshold and $\bar{\gamma}$ is the average received SNR of each transmission round. Therefore, the average BER is given by

$$\begin{aligned} P_b &= \int_{\Gamma}^{\infty} \hat{P}_b \frac{1}{\bar{\gamma}} e^{-\frac{\gamma_{\text{tot}}-\Gamma}{\bar{\gamma}}}\, d\gamma_{\text{tot}} \\ &= \int_{\Gamma}^{\infty} KQ\left(\sqrt{\eta\gamma_{\text{tot}}}\right)\frac{1}{\bar{\gamma}} e^{-\frac{\gamma_{\text{tot}}-\Gamma}{\bar{\gamma}}}\, d\gamma_{\text{tot}}. \end{aligned} \tag{131}$$

By Craig's formula,

$$Q(x) = \frac{1}{\pi}\int_0^{\pi/2}\exp\left(-\frac{x^2}{2\sin^2\theta}\right)d\theta, \tag{132}$$

we have

$$\begin{aligned} P_b &= \frac{K}{\pi\bar{\gamma}} e^{\frac{\Gamma}{\bar{\gamma}}}\int_{\Gamma}^{\infty}\int_0^{\pi/2}\exp\left(-\frac{\eta\gamma_{\text{tot}}}{2\sin^2\theta}-\frac{\gamma_{\text{tot}}}{\bar{\gamma}}\right)d\theta\, d\gamma_{\text{tot}} \\ &= \frac{K}{\pi\bar{\gamma}} e^{\frac{\Gamma}{\bar{\gamma}}}\int_0^{\pi/2}\int_{\Gamma}^{\infty}\exp\left(-\left(\frac{\eta}{2\sin^2\theta}+\frac{1}{\bar{\gamma}}\right)\gamma_{\text{tot}}\right)d\gamma_{\text{tot}}\, d\theta \\ &= \frac{K}{\pi}\int_0^{\pi/2}\frac{\exp\left(-\frac{\eta\Gamma}{2\sin^2\theta}\right)}{1+\frac{\eta\bar{\gamma}}{2\sin^2\theta}}d\theta. \end{aligned} \tag{133}$$

Next, define

$$b = \frac{\eta\Gamma}{2}, \qquad c = \frac{\eta\bar{\gamma}}{2}. \tag{134}$$

Then

$$\begin{aligned} P_b &= \frac{K}{\pi}\int_0^{\pi/2}\frac{\sin^2\theta}{\sin^2\theta+c}\exp\left(-\frac{b}{\sin^2\theta}\right)d\theta \\ &= \frac{K}{\pi}\int_0^{\pi/2}\exp\left(-\frac{b}{\sin^2\theta}\right)d\theta - \frac{Kc}{\pi}\int_0^{\pi/2}\frac{\exp\left(-\frac{b}{\sin^2\theta}\right)}{\sin^2\theta+c}d\theta. \end{aligned} \tag{135}$$

Using the integral identities

$$\frac{1}{\pi}\int_0^{\pi/2}\exp\left(-\frac{b}{\sin^2\theta}\right)d\theta = Q\left(\sqrt{2b}\right), \tag{136}$$

and

$$\frac{c}{\pi}\int_0^{\pi/2}\frac{\exp\left(-\frac{b}{\sin^2\theta}\right)}{\sin^2\theta+c}d\theta = \sqrt{\frac{c}{c+1}}e^{\frac{b}{c}}Q\left(\sqrt{2b\left(1+\frac{1}{c}\right)}\right), \tag{137}$$

it follows that

$$P_b = K\left(Q\left(\sqrt{2b}\right) - \sqrt{\frac{c}{c+1}}e^{\frac{b}{c}}Q\left(\sqrt{2b\left(1+\frac{1}{c}\right)}\right)\right). \tag{138}$$

Substituting $b = \eta\Gamma/2$ and $c = \eta\bar{\gamma}/2$ yields

$$P_b = K\left(Q\left(\sqrt{\eta\Gamma}\right) - \sqrt{\frac{\eta\bar{\gamma}}{\eta\bar{\gamma}+2}}e^{\frac{\Gamma}{\bar{\gamma}}}Q\left(\sqrt{(\eta\bar{\gamma}+2)\frac{\Gamma}{\bar{\gamma}}}\right)\right). \tag{139}$$

This completes the proof.

J AO-SCA Algorithm for (P5)

To facilitate the algorithm design, we rewrite the bandwidth allocation variables into the equivalent number of transmissions. Therefore, optimizing the bandwidth allocation $\{B_{i,j,k}\}$ is equivalent to optimizing $\{T_{i,j}\}$ under the corresponding deadline-induced symbol constraints. In particular, the cumulative deadline constraints can be written as

$$\sum_{i=1}^{k}\sum_{j=1}^{D_i} c_{i,j}N_{i,j}T_{i,j} \leq C_k, \quad k = 1,\ldots,m, \tag{140}$$

where C_k denotes the maximum number of available symbols before deadline t_k.

For the digital transmissions under Rayleigh channels, the average received SNR and the stopping threshold can be rewritten as

$$\bar{\gamma}_{i,j} = \rho_i P_{i,j}, \qquad \Gamma_{i,j} = (T_{i,j}-1)\rho_i P_{i,j}, \tag{141}$$

where $\rho_i = \sigma_{h_i}^2/\sigma_i^2$. Substituting (141) into the expected BER expression yields

$$\begin{aligned} P_{b,i,j}(T_{i,j},P_{i,j}) = K_{i,j}\Bigg[&Q\left(\sqrt{\eta_{i,j}\rho_i P_{i,j}(T_{i,j}-1)}\right) \\ &- \sqrt{\frac{\eta_{i,j}\rho_i P_{i,j}}{\eta_{i,j}\rho_i P_{i,j}+2}}e^{T_{i,j}-1}Q\left(\sqrt{(\eta_{i,j}\rho_i P_{i,j}+2)(T_{i,j}-1)}\right)\Bigg]. \end{aligned} \tag{142}$$

The resulting problem is still difficult to solve directly because $T_{i,j}$ and $P_{i,j}$ are coupled in the expected BER expression. We therefore adopt an alternating optimization framework. At the n-th outer iteration, the transmission-time variables are first optimized with the power allocation fixed at $\mathbf{P}^{(n)}$, and then the power allocation

is optimized with the updated transmission-time variables fixed at $\mathbf{T}^{(n+1)}$. Given the power allocation $\mathbf{P}^{(n)}$, the transmission-time subproblem is defined as

$$\text{(P5-T)} \quad \min_{\mathbf{T}} \quad \frac{1}{\sum_{i=1}^{m} r_i} \sum_{i=1}^{m} \sum_{j=1}^{D_i} r_i \alpha_{i,j} \beta_{i,j} \zeta_{i,j} P_{b,i,j}(T_{i,j}, P_{i,j}^{(n)}) \quad (143a)$$

$$\text{s.t.} \quad \sum_{i=1}^{k} \sum_{j=1}^{D_i} c_{i,j} N_{i,j} T_{i,j} \le C_k, \quad k = 1, \ldots, m, \quad (143b)$$

$$\sum_{i=1}^{m} \sum_{j=1}^{D_i} c_{i,j} N_{i,j} \left(P_{i,j}^{(n)} - \bar{P} \right) T_{i,j} \le 0, \quad (143c)$$

$$T_{i,j} \ge 1, \quad \forall i, j. \quad (143d)$$

The objective of P5-T is still nonconvex due to the complicated dependence of the expected BER on $T_{i,j}$. To simplify notation, we omit the indices (i, j) and write the BER term as $P_b(T, P)$. With P fixed, the first-order approximation of $P_b(T, P)$ at the current SCA point $T^{(\ell)}$ is

$$\widetilde{P}_b^T(T) = P_b(T^{(\ell)}, P) + \left. \frac{\partial P_b(T, P)}{\partial T} \right|_{T=T^{(\ell)}} \left(T - T^{(\ell)} \right). \quad (144)$$

This approximation is locally tight at $T^{(\ell)}$, since it matches both the function value and the first-order derivative of the original BER term. Replacing $P_b(T, P)$ in P5-T with $\widetilde{P}_b^T(T)$ yields a tractable surrogate problem for updating $\mathbf{T}$.

Given the updated transmission-time allocation $\mathbf{T}^{(n+1)}$, the power-allocation subproblem is defined as

$$\text{(P5-P)} \quad \min_{\mathbf{P}} \quad \frac{1}{\sum_{i=1}^{m} r_i} \sum_{i=1}^{m} \sum_{j=1}^{D_i} r_i \alpha_{i,j} \beta_{i,j} \zeta_{i,j} P_{b,i,j}(T_{i,j}^{(n+1)}, P_{i,j}) \quad (145a)$$

$$\text{s.t.} \quad \sum_{i=1}^{m} \sum_{j=1}^{D_i} c_{i,j} N_{i,j} T_{i,j}^{(n+1)} P_{i,j} \le \bar{P} \sum_{i=1}^{m} \sum_{j=1}^{D_i} c_{i,j} N_{i,j} T_{i,j}^{(n+1)}, \quad (145b)$$

$$P_{i,j} \ge 0, \quad \forall i, j. \quad (145c)$$

Similarly, with T fixed, the first-order approximation of the BER term at the current SCA point $P^{(\ell)}$ is

$$\widetilde{P}_b^P(P) = P_b(T, P^{(\ell)}) + \left. \frac{\partial P_b(T, P)}{\partial P} \right|_{P=P^{(\ell)}} \left(P - P^{(\ell)} \right). \quad (146)$$

Replacing $P_b(T, P)$ in P5-P with $\widetilde{P}_b^P(P)$ gives the corresponding surrogate problem for updating $\mathbf{P}$. The updated solution is then used as the expansion point in the next iteration.

The first-order approximations used in the SCA steps are locally tight, i.e., they match both the function value and the first-order derivative of the original BER expression at the current expansion point. Since P5 is nonconvex, the obtained solution is not guaranteed to be globally optimal. Instead, the AO-SCA procedure provides an efficient suboptimal feasible solution for the digital Rayleigh case.

K Experimental details

Table 1 summarizes the model–dataset pairs used for the seven UE requests. The accuracy denotes the error-free inference accuracy before wireless transmission distortion. The parameter number is measured in millions and is used to determine the transmission workload of each UE.

Algorithm 2 AO–SCA Algorithm for Digital Transmission over Rayleigh Fading

Require: Layer coefficients $\{\alpha_{i,j}, \beta_{i,j}, \zeta_{i,j}\}$, modulation parameters $\{K_{i,j}, \eta_{i,j}\}$, channel statistics $\{\rho_i\}$, system constraints, maximum number of iterations $I_{\max}$, and convergence tolerance ϵ.
Ensure: Feasible allocation $\{\mathbf{T}, \mathbf{P}\}$.

```
Initialize a feasible solution {T^(0), P^(0)}.
Compute the initial objective value F^(0) of (P5).
Set n = 0.
while n < I_max do
    Given P^(n), solve (P5-T) by SCA and obtain T^(n+1).
    Given T^(n+1), solve (P5-P) by SCA and obtain P^(n+1).
    Compute the objective value F^(n+1) of (P5).
    if |F^(n+1) − F^(n)| / F^(n) ≤ ε then
        break
    end if
    n ← n + 1.
end while
return {T^(n+1), P^(n+1)}.
```

Table 1: Model–dataset pairs used in the experiments.

UE	Model	Dataset	Params (M)	Acc. (%)
1	ResNet-18	CIFAR-10	11.16	93.00
2	ResNet-18	CIFAR-100	11.21	68.37
3	ResNet-18	ImageNet	11.68	70.02
4	VGG-16	ImageNet	138.34	71.25
5	ShuffleNetV2	CIFAR-100	0.44	67.81
6	MobileNetV2	CIFAR-10	2.22	94.05
7	MobileNetV2	CIFAR-100	2.33	74.31